

CE 510: Modelling and Simulation in Water Resources Engineering

Course Code: CE 510

Course Name: Modelling and Simulation in Water Resources Engineering

L-T-P-C: 2-0-2-3

Prerequisite: CE 303, Water Resources Engineering

Students intended for: B.Tech. 3rd/4th year/PG

Elective or Core: Elective

Approval: 15th Senate

Course content

- **Introduction**

Principles of simulation, nature and role of simulation in water resources engineering. Systems, models and simulation, Classification of simulation models: discrete and continuous simulation models, black box models, conceptual models, lumped and distributed models Steps involved in developing simulation models demonstrated through simple water balance model - model identification, parameter estimation - least square method for hydrologic models, brute force method, calibration and validation. [6 Lectures]

- **Water shed simulation**

Watershed and delineation; watershed characteristics; Lumped hydrologic model: HyMOD- Model conceptualization, calibration and validation Distributed hydrologic model: Simulation by physically based models - St. Venant equations; HySIM - Model conceptualization, calibration and validation [4 Lectures]

- **Urban run-off models**

Introduction to urban hydrology; Developing models for deriving IDF curve, runoff calculation and drainage network design; Storm water drainage system simulation using SWMM model including conceptualization, calibration and validation [4 Lectures]

- **Data driven models**

Black box models in hydrology; Developing artificial neural network based rainfall runoff models, model calibration and validation [4 Lectures]

- **Reservoir simulation models**

Reservoir operation policy; Developing models for determination of reservoir storage capacity, reservoir operation- deriving operational rule curves [6 Lectures]

- **Uncertainty analysis**

Monte Carlo simulation, Perturbation method, bootstrap method, first order uncertainty analysis (FOUA). [4 Lectures]

Lab Sessions

- Developing simple water balance model (Calibration- Validation) (4 Hours)
- Calibration of Hymod model parameters (3 Hours)
- Developing models for storm water drainage network (4 Hours)

- Developing ANN models, calibration and validation (4 Hours)
- Developing simulation models to fix reservoir capacity (3 Hours)
- Developing simulation models for deriving rule c curves of rese rvoir (4 Hours)
- Bootstrap methods (3 Hours)
- FOAU and Perturbation methods (3Hours)

Text Books

1. Loucks, D.P. and Eelco van Beek , **Water Resources Systems Planning and Management- an introduction to methods, models and applications**, Studies and Reports in Hydrology, UNESCO Pub., 2005.
2. Rajasekaran Pai S, G. A Vijayalakshmi, **Neural Networks, Fuzzy Logic, and Genetic Algorithms: Synthesis and Applications**, PHI Learning, 2004.

Reference Books:

1. Yeou-Koung Tung, Ben-Chie Yen, **Reliability and Uncertainty Analyses in Hydraulic Design**, ASCE Publication, 1993.
2. Averill Law, **Simulation modelling and analysis**, 4th Edition, McGraw Hill Education, 2017.
3. The manuals for hydrologic models HySIM, HyMOD and SWMM can be referred and the same will be provided during the tutorial session.
4. Rudra Pradap, **etting Started with Matlab: A Quick Introduction for Scientists and Engineers**, Oxford University Press, 2010.
5. Chow, V.T., Maidment, D.R. and Mays, L.W. **Applied Hydrology**, McGraw Hill International Editions, 1998.